

Two Dimensions of Evolution: Horizontal Within-Level and Vertical Across-Level Evolution in the Coordination Knowledge Substrate Pattern

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), the second paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its sole contribution is to articulate, in operational form, the two-dimensional evolution framework — horizontal within a structural level and vertical across structural levels — that Paper 2 specifies as the form evolution takes within the Self, and to make that framework's distinctiveness from biology's deep-time vertical evolution and from conventional one-dimensional AI-system evolution architecturally explicit.

Abstract

Paper 2 of the Coordination Knowledge Substrate (CKS) theory series introduces *bidirectional evolution*: evolution within the Self operates simultaneously on two orthogonal dimensions. Horizontal evolution is content evolution within an existing structural arrangement — DNA, action, and instinct evolving within currently composed cells, aspects, and Selves. Vertical evolution is structural reorganization — the composition relationships between levels themselves changing through governed reorganization. Both dimensions are exercised through the three evolution mechanisms (instinct, DNA, action-feedback) under human governance. Biology has both dimensions but vertical evolution requires deep time because biological structural roles are intrinsic; CKS structural roles are relational and purpose-defined, which is what makes vertical evolution available on operational timescales. This note formalizes bidirectional evolution as a standalone foundational commitment, articulates what makes it architecturally distinctive, identifies its inheritance from Paper 1 and composition with prior Phase B1 commitments, names its limits, and provides an operational test.

1. Why bidirectional evolution requires standalone formalization

The bidirectional-evolution commitment closes the dynamic-primitives portion of Phase B1. The lifecycle notes (B1.09 birth, B1.10 mating, B1.11 death) name the operations through which cells, aspects, and Selves come into existence, combine, and retire. The evolution-mechanism notes (B1.12 the three mechanisms as integrating frame, B1.13 instinct evolution as undirected mutation, B1.14 DNA evolution as directed selection, B1.15 action-feedback evolution) name the mechanisms through which already-existing entities change. Together these clusters specify *what changes and how*. They do

not yet specify *along which dimensions* evolution operates.

Bidirectional evolution names the dimensions. It is the commitment that the Self evolves in two dimensions simultaneously: across the entities at a given structural level (horizontal), and across the relational arrangements between levels (vertical). Many AI architectures provide some form of within-system evolution — model retraining, prompt iteration, knowledge-base refinement, workflow-definition revision — but these are horizontal evolution at one or another layer. Few make structural reorganization itself an architecturally available primitive. Paper 2 does, and the bidirectional-evolution commitment is what makes that fact explicit.

2. The commitment

In the CKS pattern, evolution within the Self operates on two orthogonal dimensions. Both are exercised through the three evolution mechanisms named in B1.12, and both are governed by humans per A1.01.

Horizontal evolution is content evolution within an existing structural arrangement. The composition of cells into aspects, of aspects into the Self, and the cross-level access patterns between them remain fixed; what evolves is the content carried at each level. At cell scope, DNA-layer substrates evolve through directed selection (B1.14), the underlying instinct layer sharpens through governed integration of capability jumps (B1.13), and accumulated action-layer recordings drive DNA refactoring (B1.15). At aspect scope, the orchestration substrates governing coordination among constituent cells evolve through directed selection of aspect-level rules and through action-feedback proposals based on accumulated aspect operations. At Self scope, the integration architecture holding multiple aspects as one whole evolves through directed selection of Self-level specifications and through accumulated cross-aspect experience. The hallmark is that the *what-composes-what* relationship between levels does not change.

Vertical evolution is structural reorganization. The composition relationships between levels themselves change through governed reorganization. Specifically:

- *Cell reassignment.* A cell that participates in one aspect can be reassigned to another, or to multiple aspects simultaneously, by changing the relational role assignments per B1.17 (forthcoming).
- *Aspect splitting.* An aspect whose purposes have diverged can split into two or more, redistributing constituent cells and refactoring coordination substrates per the new purpose boundaries.
- *Aspect merging.* Aspects whose purposes have converged can merge, combining their coordination substrates per the unified purpose.
- *Aspect dissolution.* An aspect whose purpose is no longer needed can be dissolved, releasing or reassigning its constituent cells.
- *Aspect introduction.* A new aspect can be introduced when a new purpose emerges, drawing constituent cells from existing aspects and possibly originating new cells through birth (B1.09).
- *Cross-level access pattern reconfiguration.* The pattern by which higher levels access lower levels can be reconfigured per B1.19 (forthcoming).

Paper 2 makes this explicit: the structural arrangement is itself substrate content (orchestration substrates that define composition) and is human-governed in the same way DNA-layer content is. Vertical evolution operates on that substrate content through the same three mechanisms that drive horizontal evolution.

The two dimensions are orthogonal — vertical reorganization does not pause horizontal refinement, and horizontal refinement does not preclude future vertical reorganization. Both record evolution events with the provenance metadata named in A2.40 (forthcoming), preserving path retraceability per A1.07. Both operate within the Self per Paper 2's scope statement; inter-Self dynamics are out of scope.

3. What makes bidirectional evolution architecturally distinctive

Two contrasts make the distinctiveness clear: with conventional one-dimensional AI-system evolution, and with biological evolution.

Contrast with conventional one-dimensional evolution. Most AI architectures provide a single evolution dimension. Model-centric architectures evolve through retraining or fine-tuning. Knowledge-base-centric architectures evolve through content additions, deletions, and edits. Workflow-engine-centric architectures evolve through workflow-definition revisions. In each case the evolution is *within an existing structural arrangement*; the architectural arrangement itself is fixed at design time and reorganized only through external system replacement. The CKS bidirectional commitment specifies two dimensions across three structural levels, producing six intersections of dimension and level. The structural arrangement is itself evolvable substrate content rather than a fixed design-time specification.

Contrast with biological evolution. Biology has both dimensions but on radically different timescales. Within-level evolution proceeds tractably — cells turn over within an organism's lifespan, populations evolve across generations. Vertical evolution — body-plan reorganization, the emergence of new tissue types, the introduction of new organ systems — requires deep evolutionary time. The Cambrian explosion, the evolution of multicellularity, the emergence of nervous systems each took millions of years. Within a given species, the body plan is essentially fixed.

The reason is structural. Biological cells are *intrinsically* committed to their tissue and organ memberships through development. A mature cardiac myocyte cannot be reassigned to function as a hepatocyte. The only way to change body-plan structure is across generations through evolutionary modification of developmental programs.

CKS exceeds biology at this point — explicitly named in Paper 2's *Where CKS exceeds biology* section as *structural evolvability on operational timescales* and *relational rather than intrinsic structural roles*. CKS cells are not intrinsically committed to particular aspects. Their participation is a relational role assignment named in orchestration substrate, and reassignment is a substrate change governed by the same authority architecture that governs DNA changes. The CKS Self can therefore restructure on operational timescales — within the lifetime of a single deployment — through governed reorganization decisions, not across generations through deep evolutionary time. The biological analog functions as conceptual scaffold; the architectural substance is the two-dimensional evolution framework operating on shared timescales under unified human governance.

4. Inheritance and composition

Bidirectional evolution does not introduce new axioms. It composes inherited Paper 1 commitments with prior Phase B1 commitments.

Inherited from Paper 1. A1.01 (human-governed: authority, not labor) is exercised in both dimensions: the rights to inspect, modify, and override apply both to content carried at each level and to the orchestration substrates that define composition. A1.07 (path retraceability) is preserved through evolution events recorded with the six-field provenance metadata named in A2.40 (forthcoming). A2.04 (orchestration rule authoring as governance, forthcoming) is the foundation for both dimensions: humans author rules specifying horizontal evolution patterns per level and vertical reorganization patterns per restructuring type. A2.47 (authority distribution, forthcoming) and the boundary cases A6.06, A6.12, A6.14 (all forthcoming) apply when vertical reorganization spans multiple aspects, multiple authors, or accumulated rule versions across deployment lifetimes.

Composed with prior Phase B1 commitments. Bidirectional evolution depends on B1.02 (three-level structure: cell, aspect, Self) — without three levels there is no across-level dimension to evolve along. B1.03–B1.05 (cell, aspect, Self elaborations) supply the objects of horizontal evolution and the participants in vertical reorganization, with aspects being the primary site of vertical reorganization. The lifecycle notes (B1.09 birth, B1.10 mating, B1.11 death) apply at both dimensions: birth originates entities horizontally and originates new aspects vertically; mating combines existing entities and combines aspect compositions; death retires entities and dissolves aspects. The mechanism notes (B1.12 frame, B1.13–B1.15 mechanisms in detail) supply the engines for both dimensions — vertical evolution is most prominently driven through directed selection (humans authoring reorganization rules), but action-feedback can prompt vertical-reorganization proposals when accumulated experience reveals that the existing arrangement no longer fits its purpose. B1.17 (relational and purpose-defined structural roles, forthcoming) is what makes vertical evolution operationally available — if structural roles were intrinsic, vertical evolution could not operate on operational timescales.

The composition is consistent with the recursive Paper 1 commitments principle (B1.20, forthcoming): Paper 1's commitments hold at each level of the three-level structure, and bidirectional evolution operates on them at each level under the same authority architecture.

5. Operational implications

Both dimensions are configured per deployment. A deployment configures horizontal evolution rules per level and vertical reorganization rules per restructuring pattern. The architecture provides the framework; the rules are deployment-specific authored substrate content.

Horizontal evolution is the steady-state mode; vertical evolution is event-driven. Day-to-day evolution proceeds through horizontal refinement: cells refine DNA based on accumulated action layers, aspects update coordination rules, Selves refactor integration based on cross-aspect experience. Vertical reorganization occurs less frequently, when restructuring serves a purpose that horizontal refinement cannot achieve — when an aspect's purposes have genuinely diverged, when a new purpose has emerged the existing structure cannot accommodate, when an aspect's purpose has been retired.

Vertical evolution often involves multiple cells or aspects simultaneously. Splitting an aspect involves redistributing constituent cells; merging combines coordination substrates and resolves rule conflicts via Paper 1's conflict-as-first-class-object commitment; introducing a new aspect involves drawing cells from existing aspects through reassignment and possibly originating new cells through birth. Cross-aspect vertical evolution composes with the authority-distribution boundary cases (A2.47, A6.06, A6.12) when authority spans multiple human authors.

Deployments may emphasize one dimension over the other. A high-throughput, stable-purpose deployment may emphasize horizontal refinement and rarely exercise vertical reorganization. A deployment in a period of organizational restructuring may emphasize vertical reorganization to align Self structure with changing organizational structure (the *structural co-adaptation* property named in Paper 2's enterprise-application extension claim). Both emphases are admissible.

6. What bidirectional evolution is NOT

Not without governance. Both dimensions are human-governed per A1.01. Vertical reorganization is not autonomous — the commitment is that vertical evolution is *available*, not that it is *automatic*.

Not automatic. Neither dimension proceeds without deliberate human action through authored rules. Action-feedback may surface proposals, but adoption is governed.

Not inter-Self. Bidirectional evolution operates within the Self per Paper 2's scope statement.

Not eliminating level distinctions. Vertical reorganization changes the relational arrangement between cells, aspects, and Selves; it does not erase the level distinctions named in B1.02. Cells remain cells, aspects remain aspects, Selves remain Selves; what changes is which entities participate in which arrangements.

Not bypassing retroactivity treatment. Evolution does not retroactively alter past substrate state. A6.02 (rule retroactivity treatment, forthcoming) applies — evolution events produce new substrate content alongside preserved historical state. A cell reassigned from aspect A to aspect B retains its history of participation in aspect A as recorded substrate; the reassignment is a new event, not a rewriting of the past.

Not requiring both dimensions to be active simultaneously. The commitment is architectural availability, not operational simultaneity. A deployment that performs only horizontal evolution and never exercises vertical reorganization satisfies the commitment.

Not prescribing organizational structure. The horizontal/vertical framing names dimensions of evolution, not particular structures. The geometric framing is conceptual scaffold; the architectural substance is the within-level versus across-level distinction.

7. Operational test

A system instantiates the bidirectional-evolution commitment if and only if all of the following are true:

1. Evolution within the Self operates on two distinguishable dimensions: horizontal (within a structural level) and vertical (across structural levels). Both are reachable by humans under the

governance commitment named in A1.01.

2. Horizontal evolution is exercisable at each of the three structural levels (cell, aspect, Self) per B1.02–B1.05.
3. Vertical evolution includes, as architecturally available primitives, at least: cell reassignment across aspects, aspect splitting, aspect merging, aspect dissolution, aspect introduction, and cross-level access pattern reconfiguration.
4. Both dimensions are exercised through the three evolution mechanisms per B1.12 — instinct evolution as undirected mutation, DNA evolution as directed selection, action-feedback evolution.
5. Both dimensions preserve path retraceability per A1.07: evolution events are recorded with provenance metadata per A2.40 (forthcoming) such that past substrate state remains reconstructible.
6. Structural roles are relational and purpose-defined per B1.17 (forthcoming) rather than intrinsic, enabling vertical reorganization on operational timescales rather than evolutionary timescales.
7. Both dimensions operate within the Self per Paper 2's scope statement.

A system that fails any of (1)–(7) may evolve in some other sense, but it does not instantiate the CKS bidirectional-evolution commitment.

Conclusion

Bidirectional evolution names the two-dimensional framework within which evolution operates in the CKS Self. Horizontal evolution is content evolution within an existing structural arrangement; vertical evolution is structural reorganization through governed reorg. Both are exercised through the three evolution mechanisms under human authority. The combination is architecturally distinctive in two ways: it exceeds conventional one-dimensional AI-system evolution by making structural reorganization an architecturally available primitive, and it exceeds biological evolution by operating both dimensions on operational timescales — a capability biology cannot match because biological structural roles are intrinsic, while CKS structural roles are relational and purpose-defined.

This note closes the dynamic-primitives portion of Phase B1. Subsequent Phase B1 notes shift to structural properties: relational-and-purpose-defined structural roles (B1.17), the higher-as-content-domain property (B1.18), cross-level access (B1.19), and the recursive Paper 1 commitments (B1.20).

Subsequent work that adopts the CKS pattern, extends it, composes it with adjacent patterns, or argues against it should use *bidirectional evolution* in the sense formalized here. Subsequent work that uses the term differently is using a different concept, and the difference should be named.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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